

PHARMACY MANAGEMENT SYSTEM

PROBLEM STATEMENT:

Pharmacies face challenges in managing medications due to their complexity and large number, making it difficult for pharmacists to easily update, delete, or add new medications.

INTRODUCTION:

The pharmacy management system is designed to overcome the challenges of handling and managing medications in pharmacies. This application allows administrators to log in using their username and password, and then they are transferred to the homepage. From there, they can perform a range of operations on medication data, such as adding a new medication, modifying existing medication data, viewing medications, or deleting medication. If an administrator is new and does not have an account, they can register using the "Sign In" button. The medication data table contains the medication name, ID, and prescription type, which contribute to working on the previous four operations. Every pharmacy, whether small or large, has challenges to overcome like managing medication information. This application is designed to solve this problem.

TECHNOLOGIES USED FOR IMPLEMENTATION:

Front-End:

- **Swing and AWT:**

We used the swing in most of the project and used AWT for some command in the project such as the GridLayout. We used both for designing the pages (Log in and sign in page, Home Page) and perform the operations.

Back-End:

- **JDBC:**

We used JDBC to connect to the databases, execute SQL commands, and retrieve and return data to the application.

- **Database:**

We used mySQL through is XAMPP server to create the database in our application.

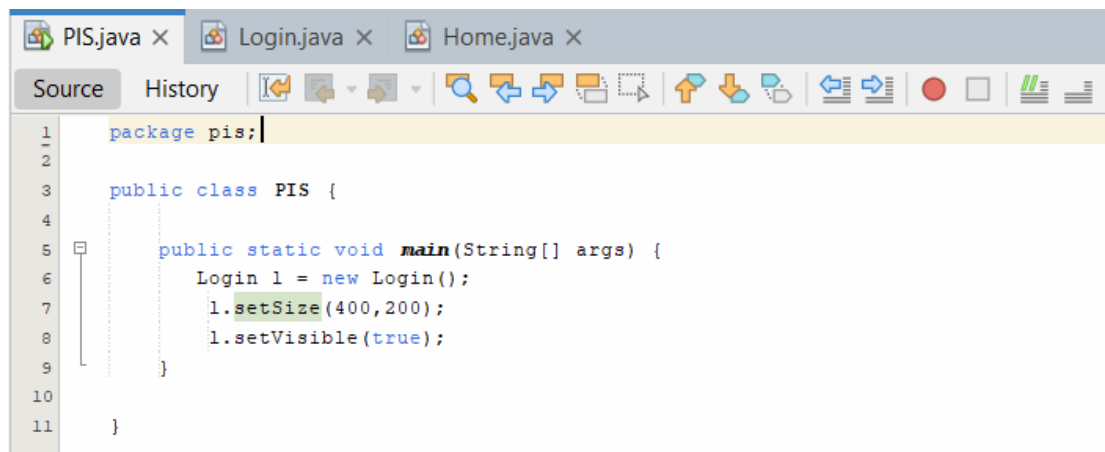
CODE:

PIS.java

```
package pis;

public class PIS {

    public static void main(String[] args) {
        Login l = new Login();
        l.setSize(400,200);
        l.setVisible(true);
    }
}
```



Login.java

```
package pis;
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import java.sql.*;
public class Login extends JFrame implements ActionListener
{
    JLabel D1 =new JLabel("User Name: ");
```

```

JTextField t1 =new JTextField(30);
JLabel D2 =new JLabel("Password: ");
JTextField t2 =new JTextField(30);
JButton b1=new JButton("Sing In");
JButton b2=new JButton("Log In");

public Login(){
    JPanel p1=new JPanel();
    p1.setLayout(new GridLayout(4,2,6,6));
    p1.add(D1);
    p1.add(t1);
    p1.add(D2);
    p1.add(t2);
    p1.add(b1);
    p1.add(b2);
    add(p1);
    b1.addActionListener(this);
    b2.addActionListener(this);
}

public void actionPerformed(ActionEvent ae){
    String UN = t1.getText();
    String Pass = t2.getText();

    if(ae.getSource()==b1)
    {

        try{
            String url ="jdbc:mysql://localhost:3306/pms";
            Connection conn = DriverManager.getConnection(url,"root","");
            Statement st = conn.createStatement();
            int a = st.executeUpdate("insert into
login(Username>Password)Values('"+UN+"','"+Pass+"')");

            if(a==1)
                JOptionPane.showMessageDialog(this,"Inserted Succesfully");
            st.close();
            conn.close();
        }

        catch(SQLException e){
            JOptionPane.showMessageDialog(null, e);

        }

    }

    if(ae.getSource()==b2)
    {
        if(checkU(UN,Pass)){

```

```

        dispose();
        Home m = new Home();
        m.setSize(400,300);
        m.setVisible(true);

    }else
        JOptionPane.showMessageDialog(null, "User Name or Password is
incorrect");
    }

}

public static boolean checkU(String un,String pass){
    boolean st = false;
    try{
        String url ="jdbc:mysql://localhost:3306/pms";
        Connection conn = DriverManager.getConnection(url,"root","");
        PreparedStatement ps = conn.prepareStatement("select * from login where
Username=? and Password=?");
        ps.setString(1,un);ps.setString(2,pass);
        ResultSet rs = ps.executeQuery();
        st=rs.next();
    }
    catch(SQLException e){
        e.printStackTrace();
    }
    return st;
}
}

```

```

1 package pis;
2 import javax.swing.*.*;
3 import java.awt.*.*;
4 import java.awt.event.*;
5 import java.sql.*;
6 public class Login extends JFrame implements ActionListener
7 {
8     JLabel D1 =new JLabel("User Name: ");
9     JTextField t1 =new JTextField(30);
10    JLabel D2 =new JLabel("Password: ");
11    JTextField t2 =new JTextField(30);
12    JButton b1=new JButton("Sing In");
13    JButton b2=new JButton("Log In");
14
15
16    public Login(){
17        JPanel p1=new JPanel();
18        p1.setLayout(new GridLayout(4,2,6,6));
19        p1.add(D1);
20        p1.add(t1);
21        p1.add(D2);
22        p1.add(t2);
23        p1.add(b1);
24        p1.add(b2);
25        add(p1);
26        b1.addActionListener(this);
27        b2.addActionListener(this);
28    }
29
30    public void actionPerformed(ActionEvent ae){
31        String UN = t1.getText();
32        String Pass = t2.getText();
33
34        if(ae.getSource()==b1)
35        {
36
37            try{
38                String url ="jdbc:mysql://localhost:3306/pms";
39                Connection conn = DriverManager.getConnection(url,"root","");
40                Statement st = conn.createStatement();
41                int a = st.executeUpdate("insert into login (Username,Password)Values ('"+UN+"','"+Pass+"')");
42
43                if(a==1)
44                {
45                    JOptionPane.showMessageDialog(this,"Inserted Succesfully");
46                    st.close();
47                    conn.close();
48                }
49            } catch (SQLException e){
50                JOptionPane.showMessageDialog(null, e);
51            }
52
53        }
54
55        if(ae.getSource()==b2)
56        {
57            if (checkU(UN,Pass)){
58                dispose();
59                Home m = new Home();
60                m.setSize(400,300);
61                m.setVisible(true);
62            }
63            else
64            {
65                JOptionPane.showMessageDialog(null, "User Name or Password is incorrect");
66            }
67        }
68    }
69
70    public static boolean checkU(String un,String pass){
71        boolean st = false;
72        try{
73            String url ="jdbc:mysql://localhost:3306/pms";
74            Connection conn = DriverManager.getConnection(url,"root","");
75            PreparedStatement ps = conn.prepareStatement("select * from login where Username=? and Password=?");
76            ps.setString(1,un);ps.setString(2,pass);
77            ResultSet rs = ps.executeQuery();
78            st=rs.next();
79        } catch (SQLException e){
80            e.printStackTrace();
81        }
82        return st;
83    }
84 }

```

Home.java

```

package pis;
import java.awt.*.*;
import javax.swing.*.*;

```

```

import java.sql.*;
import java.awt.event.*;
public class Home extends JFrame implements ActionListener{

    JLabel ml = new JLabel("Medicine Name: ");
    JLabel idl = new JLabel("Medicine ID: ");
    JLabel pl = new JLabel("Prescription Type:");

    JTextField name = new JTextField(20);
    JTextField mid = new JTextField(10);
    JTextField pres = new JTextField(10);

    JButton view = new JButton("View");
    JButton add = new JButton("Add");
    JButton update = new JButton("Update");
    JButton del = new JButton("Delete");

    public Home(){

        JPanel p1 = new JPanel();
        p1.setLayout(new GridLayout(6,2,6,6));

        p1.add(ml);
        p1.add(name);

        p1.add(idl);
        p1.add(mid);

        p1.add(pl);
        p1.add(pres);

        p1.add(view);
        p1.add(add);
        p1.add(update);
        p1.add(del);

        add(p1);
        view.addActionListener(this);
        add.addActionListener(this);
        update.addActionListener(this);
        del.addActionListener(this);
    }
    public void actionPerformed(ActionEvent ae)
    {
        String n = name.getText();
        String id = mid.getText();
        String p = pres.getText();
        int a = 0;
    }
}

```

```

try{
String url = "jdbc:mysql://localhost:3306/pms";
Connection conn = DriverManager.getConnection(url,"root","");
Statement st = conn.createStatement();

if(ae.getSource()==view)
{
    ResultSet rs = st.executeQuery("SELECT * FROM medicine");
    while(rs.next()){
        System.out.println(rs.getString(1)+ " | "+ rs.getString(2) + " | " +
rs.getString(3));
    }
}
if(ae.getSource()==add){
    a = st.executeUpdate("insert into medicine(name,id,PrescriptionType)
values(\""+n+"\",\""+id+"\",\""+p+"")");
}
if(ae.getSource()==update){
    a = st.executeUpdate("update medicine set name = \""+n+"\",PrescriptionType =
\""+p+"\" where id = \""+id+"\" ");
}
if(ae.getSource()==del){
    a = st.executeUpdate("delete from medicine where id = \""+id+"\" ");
}
if(a==1){
    JOptionPane.showMessageDialog(this, "Done Succesfully");
    st.close();
    conn.close();
}

} catch(SQLException e){
    JOptionPane.showMessageDialog(null, e);
}
}
}

```



```
1 package pis;
2 import java.awt.*;
3 import javax.swing.*;
4 import java.sql.*;
5 import java.awt.event.*;
6 public class Home extends JFrame implements ActionListener{
7
8
9     JLabel m1 = new JLabel("Medicine Name: ");
10    JLabel id1 = new JLabel("Medicine ID: ");
11    JLabel p1 = new JLabel("Prescription Type:");
12
13    JTextField name = new JTextField(20);
14    JTextField mid = new JTextField(10);
15    JTextField pres = new JTextField(10);
16
17    JButton view = new JButton("View");
18    JButton add = new JButton("Add");
19    JButton update = new JButton("Update");
20    JButton del = new JButton("Delete");
21
22    public Home(){
23
24
25        JPanel p1 = new JPanel();
26        p1.setLayout(new GridLayout(6,2,6,6));
27
28        p1.add(m1);
29        p1.add(name);
30
31        p1.add(id1);
32        p1.add(mid);
33
34        p1.add(p1);
35        p1.add(pres);
36
37        p1.add(view);
38        p1.add(add);
39        p1.add(update);
40        p1.add(del);
41
42        add(p1);
43        view.addActionListener(this);
44        add.addActionListener(this);
45        update.addActionListener(this);
46        del.addActionListener(this);
47    }
48
49    public void actionPerformed(ActionEvent ae)
50    {
51        String n = name.getText();
```



```
PIS.java x Login.java x Home.java x
Source History
47 del.addActionListener(this);
48
49 public void actionPerformed(ActionEvent ae)
50 {
51     String n = name.getText();
52     String id = mid.getText();
53     String p = pres.getText();
54     int a = 0;
55     try{
56         String url = "jdbc:mysql://localhost:3306/pms";
57         Connection conn = DriverManager.getConnection(url,"root","");
58         Statement st = conn.createStatement();
59
60         if(ae.getSource()==view)
61         {
62             ResultSet rs = st.executeQuery("SELECT * FROM medicine");
63             while(rs.next()){
64                 System.out.println(rs.getString(1)+ " | " + rs.getString(2) + " | " + rs.getString(3));
65             }
66         }
67         if(ae.getSource()==add){
68             a = st.executeUpdate("insert into medicine(name,id,PrescriptionType) values('"+n+"','"+id+"','"+p+"')");
69         }
70         if(ae.getSource()==update){
71             a = st.executeUpdate("update medicine set name = '"+n+"',PrescriptionType = '"+p+"' where id = '"+id+"' ");
72         }
73         if(ae.getSource()==del){
74             a = st.executeUpdate("delete from medicine where id = '"+id+"' ");
75         }
76         if(a==1){
77             JOptionPane.showMessageDialog(this, "Done! Succesfully");
78             st.close();
79             conn.close();
80         }
81     }catch(SQLException e){
82         JOptionPane.showMessageDialog(null, e);
83     }
84 }
85 }
```

Database

Databases

Create database

Create new table

Table name: Number of columns:

Table name: Add column(s)

Name	Type	Length/Values	Default	Collation	Attributes	Null	Index
Username	VARCHAR	50	None			☐	...
Click from Central Columns							
Password	VARCHAR	30	None			☐	...
Click from Central Columns							

Table comments:

Collation:

Storage Engine:

PARTITION definition:

Partition by: (Expression or column list)

Partitions:

Create new table

Table name

Number of columns

medicine

3

Create

Table name: Add column(s)

Name	Type	Length/Values	Default	Collation	Attributes	Null	Index
name Pick from Central Columns	VARCHAR	50	None			☐	---
id Pick from Central Columns	VARCHAR	20	None			☐	---
PrescriptionType Pick from Central Columns	VARCHAR	60	None			☐	---

Table comments:

Collation:

Storage Engine:

PARTITION definition:

Partition by: (Expression or column list)

Partitions:

INPUT AND OUTPUT:

Sign in:

User Name:

Password:

Sing In

Log In

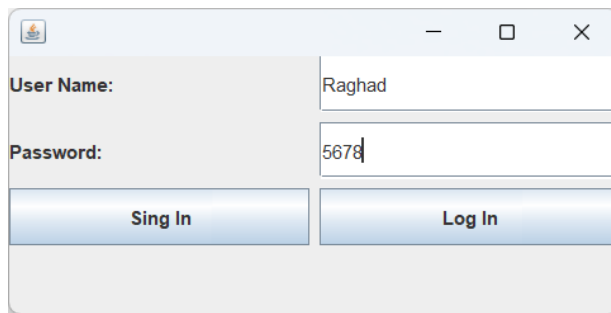
Message

Inserted Succesfully

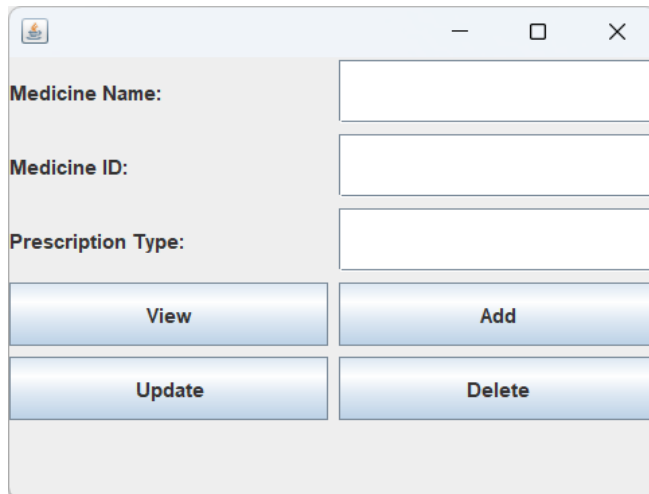
OK

Username	Password
Aryam	1234
Raghad	5678

Log in:

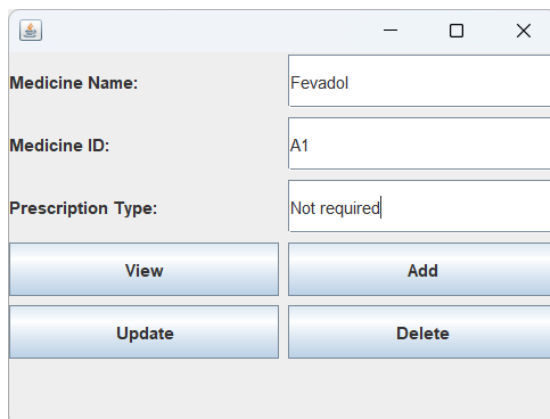


A login window with a title bar containing a small icon and standard window controls. It features two input fields: 'User Name:' with the text 'Raghad' and 'Password:' with the text '5678'. Below the fields are two buttons: 'Sing In' and 'Log In'.

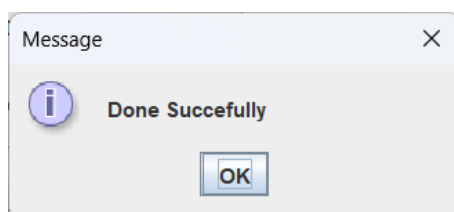


A window for managing medicines with a title bar. It contains three input fields: 'Medicine Name:', 'Medicine ID:', and 'Prescription Type:'. Below these fields are four buttons arranged in a 2x2 grid: 'View', 'Add', 'Update', and 'Delete'.

Add:



An 'Add' window for medicines with a title bar. It contains three input fields: 'Medicine Name:' with the text 'Fevadol', 'Medicine ID:' with the text 'A1', and 'Prescription Type:' with the text 'Not required'. Below these fields are four buttons arranged in a 2x2 grid: 'View', 'Add', 'Update', and 'Delete'.



A 'Message' dialog box with a title bar. It displays an information icon, the text 'Done Succesfully', and an 'OK' button.

name	id	PrescriptionType
Fevadol	A1	Not required

Another add:

Medicine Name:

Tabuvan

Medicine ID:

A2

Prescription Type:

From doctor

View

Add

Update

Delete

View:

- > q4
- > Q55-1.0-SNAPSHOT
- > SQ5-1.0-SNAPSHOT

Medicine Name:

Fevadol

Medicine ID:

A1

Prescription Type:

Not required

View

Add

Update

Delete

Output - PIS (run)

```
run:
Fevadol | A1 | Not required
```

Another view (after adding and Update):

```
Fevadol | A1 | Not required
Fevadol | A1 | No prescription required
Tabuvan | A2 | From doctor
```

View

Add

Update

Delete

Update:

Medicine Name:

Fevadol

Medicine ID:

A1

Prescription Type:

No prescription required

View

Add

Update

Delete

Message

i

Done Succefully

OK

name	id	PrescriptionType
Fevadol	A1	No prescription required

Delete:

Medicine Name:

Medicine ID:

A1

Prescription Type:

View

Add

Update

Delete

Message

i

Done Succefully

OK

name	id	PrescriptionType
------	----	------------------

Tabuvan A2 From doctor

Last view:

Tabuvan	A2	From doctor	View	Add
			Update	Delete